

Smart Air Purifier

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Abstract— In today's society, air pollution is an ever-lasting environmental problem. Automobile emissions, factory emissions, smoke from the cigarette and many other factors are the leading cause of pollution. This approach can be viewed as a preventative measure to control air pollution at home. The goal of this paper is to introduce a Smart Air Purifier. The sensors which are attached to it is Arduino Uno monitor the level of pollutants in the Environment. If the sensed value goes beyond the program's set value, which is the threshold value, the fan will start moving and air suction will take place inside the box which will take air from the surroundings and then with the help of Hepa filter and fan the purified air will send out. Till the room value of pollutants from the smoke gets lower than the set value the device will work and after the lower value is achieved the device will stop. Finally, the room air will get cleaned and it would be safe to inhale. The best part of this paper is that it can be carried anywhere because of its compact design.

Keywords— Air Pollutants Control, Engine Hepa Filter, Arduino Uno.

I. INTRODUCTION

Ecological contamination within the country has been a solemn problem of the current century. The primary source of contamination in India is because of car motors, Factories' smoke, gases from the Air Conditioner, and the burning of waste. Authorities of India made many guidelines to control ecological contamination triggered because of vehicle discharge, and factory smoke but most of them turn out to be abortive. In many Indian states, air quality lies above the 300 marks. in Delhi it is 389, then in Patna, it is 353 then in Greater Noida it is 350, and after that it is Hyderabad, it is 233. These data are dated 27-10-2020 which is issued by the Government of India. As the report says the Air quality which is above the 250 is very unhealthy and as the data predicted many of the metropolitan cities data is more than 250 which is very unhealthy to breathe especially for children as they will inhale that and that's not good for their health same goes for the adult people inhaling these type of air can have a serious effect on the lungs [9].

Carbon monoxide lowers blood's ability to transport oxygen, and hydrocarbons in the atmosphere have an impact on the heart, brain, kidneys, and bone marrow. Nitrogen oxide irritates the lungs, resulting in respiratory problems. It's tough to evaluate every case of air pollution independently during a time of urbanization due to the fast rise of Industrialization and the motor. Inspection of every cause necessitates a significant amount of Manpower and more money.

To keep an eye on all the air quality we breathe in, we introduced a system known as Smart Air Purifier, with the help of which we can keep an eye on the air we inhale with very ease. It has monitored that help indicates the air quality of the room which helps us to keep in check with the air index. Sensors are a vibrant part of this paper; the sensors help to indicate the value of the various pollutants in the air which easily helps the fan how long they have to operate.

Research paper have following structure: -

- Literature Survey
- Proposed Methodology
- Workflow analysis
- Result and discussion
- Conclusion

II. LITERATURE SURVEY

In past, a lot of research papers have been published on the same problem but they did not include the Hepa filter and automatic working of the device [1] use the sensor to monitor the air quality. The sensor are gas sensor MQ-135, we can also use more sensor like temperature, humidity. Once checked then the air gets filter according to different house and needs. This also help user to automatically turn on or off the device [2].

In this method there is a problem regards with the use case of the device. The user who is using this device have switch on and off the device manually. This is the bigger problem as device only tell the user that air is filter and u need to stop the device as in the night if there is a gas leakage or any bad air scenarios, they will not be able to wake up and switch on the device. And we cannot make the device work for all day long because it will consume to much of the electricity this will led to one of the bigger problems of more energy consumption so to counter that problem, we need to use our type of filter. Similarly, if we take the devices available in the market, they are much costly than our device. [4-7].

On the other hand, if we talk about the previously done work on the topic called UAV (unmanned aerial vehicle). That method will also talk about the indoor monitoring system which measures the air quality of the building [6]. This will only gather and tell about the information of the quality of the air in system.

The vehicles are very costly and if we talk about the maintenance cost that is also very high. The UAV will only gather quality of air through various sensors.

III. PROPOSED METHODOLOGY

This system uses, mq135 sensors sited at the MDF box, to monitor the carbon monoxide, nitrogen oxide, and hydrocarbon value in the air, which is measured. Then the measure-valued goes to the input of the pin of Arduino. Then the input value is processed if the value of the mq_135 Sensor is more than 100 then the LCD monitor displays the value indicating it's bad quality air, if the value input is less than 100

but if it's greater than 40 then the LCD monitor the value is indicated, and if the value is less than the 40 then the monitor will display that it's the best. When the air quality index is more than 40 then the exhaust fan starts to rotate this takes the air from the environment and then air passed through the HEPA filter where all the dust gets filtered and then filtered air gets thrown out of the fan again. If the value of air quality goes below 40 then by output pin the relay module gets into play and breaks the circuit. The output is shown on the monitor. This will help to filter the bad air from the environment. The room should be isolated i.e. closed otherwise it will go on infinite mode as the air input will not stop so the relay will not stop.

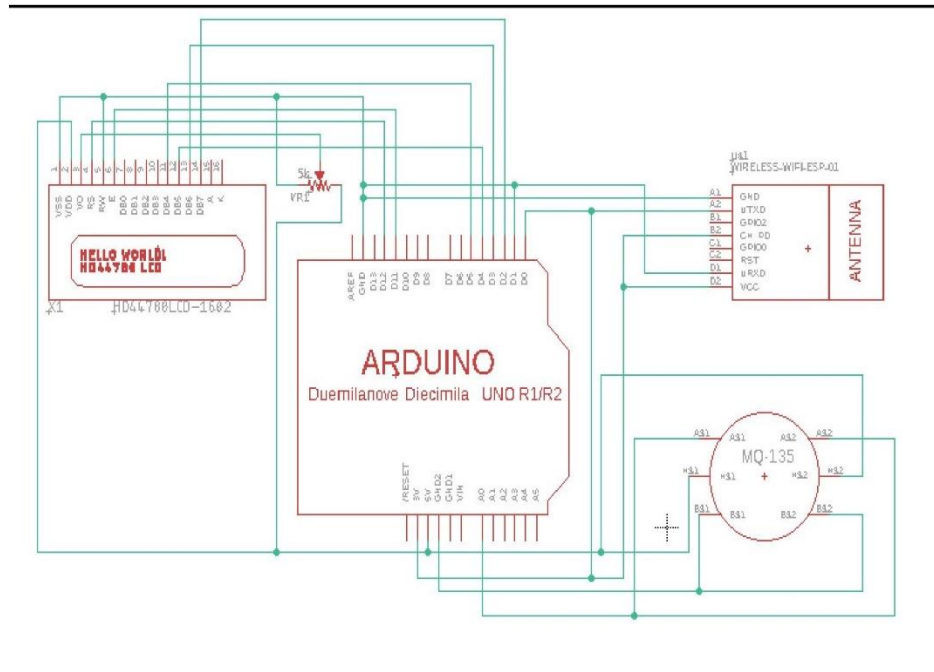


Fig. 1. Block Diagram of the Module

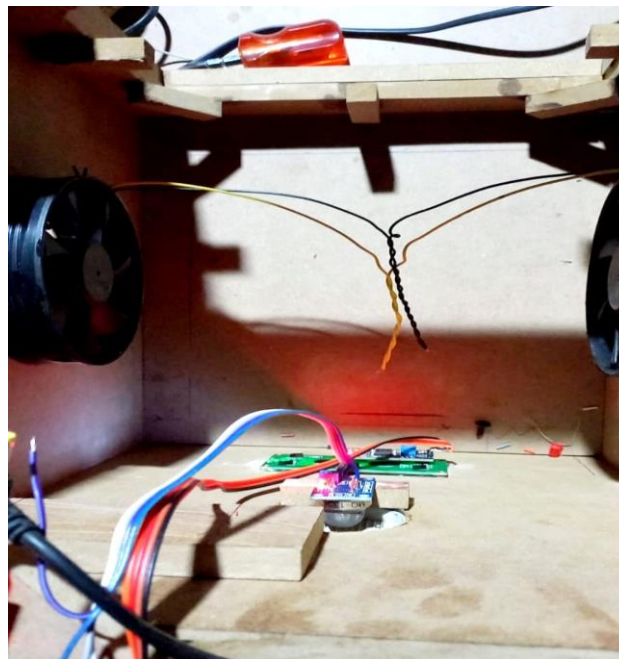


Fig. 2. Internal circuit diagram

IV. WORKFLOW ANALYSIS

The pre-filter air present in the room goes through the MQ135 sensor of the air purifier, which senses the amount of smoke, carbon dioxide, ammonia, nitrogen oxide, and other harmful elements present in the air. This results in determining an approximate amount of ppm of the air in the room. This data is transferred via the input pin to Arduino UNO, which works on about +5V and gives a digital as well as analog output for the input received through the sensor. This open-source microcontroller is programmed to work as a relay for the system, hence when the sensor detects air quality above the threshold quality set by the user, it activates the exhaust fans placed in the system, therefore resulting inflow of air present in the room through the HEPA filters. These filters thus separate the harmful elements in the air and fresh air comes out after passing through them. The use of the HEPA filter is to eliminate NO_x, NH₃, C_xH_x, and other such gases including smoke. Thus the fresh air is sent out from the device back to the room through an exhaust fan. The air coming out from the purifier is at much lower ppm and thus serves the motive of the device. This process continues till the sensor detects air quality to be lower than the set threshold, which leads to Arduino breaking the circuit and stopping the function of fans thus auto-switching off the device with no manual or external command needed.

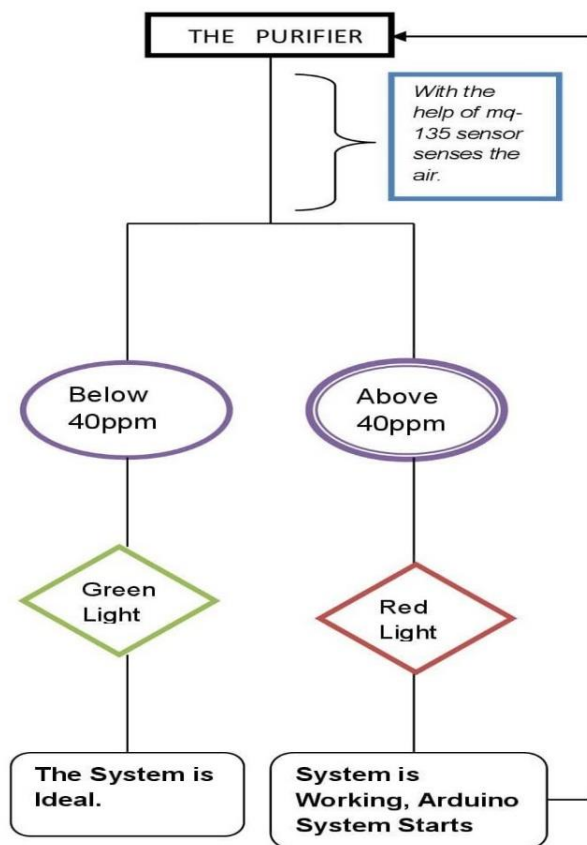


Fig. 3. Flow Chart Describing the Workflow

V. RESULTS AND DISCUSSIONS

Arduino Uno is used for the implementation of the air quality measurement system. It took three key steps to implement the complete measurement system. In the first step, the system is started and commanded by a sensor to start receiving real-time data. The second stage works to check the information with our given situation, that is, if the quality of the air is more than the given initial value, it displays a message that air filtration will be started. The third stage is air filtration, where it starts to clean the air until the air quality drops to a certain threshold value.

A. Stage – 1 (Initialize system)

The accuracy of the detectors is to accurately read the quality of air. It's the main stage for the system. An LCD will show an initialization message for the start-up and the sensors will start to take and read the air quality values. When the first value is received and given to the system by the sensor, the values of the air quality are displayed on the LCD screen. Every 1 second, the sensor takes the new value of the index and updates it to the system and displays it on the LCD. The sensors are able to find all the necessary harmful gases within the air.



Fig. 4. Reading and displaying air index quality

B. Stage-2 (Compare acquired values)

When the gas sensors receive the quality index values, it is compared with the given set value obtained with the normal index of air quality. The air quality values are given below

TABLE I. QUALITY OF AIR INDEX

Numerical value	Index of air	Effects
0-40	Best	Air quality is best
41-100	Good	Air quality can be acceptable
101-200	Moderate	Unhealthy affected on some people
>200	Hazardous	Very Unhealthy

TABLE II. SYSTEM PERFORMANCE AT DIFFERENT VALUES

Values	Action
0-50 (best)	Not working
51-100 (better)	working
101-200 (moderate)	working
>300(hazardous)	working

As the table shows, the AQI for different values affects different people differently. When we acquire the AQI values, our device compares the received values with the adjusted values. If the received AQI value is over 40, it automatically commences the filtration of air. For our device, the above-referred AQI values are taken and issued to the system.

C. Stage-3 (Purify air)

This device contains an air filter. This air filter is activated when our system receives a command from the gas sensor. This process will continue till air quality gets purified and become normal. As our system measures index value for every 1 seconds, it will make the air filter efficient.

VI. CONCLUSION

Using this method which is an efficient way of reducing and removing the air pollution caused due to the gases emitted in the domestic household caused due to the vehicles, smoke from the cigarettes. The overall air pollution and the air quality index of the room or cabin get reduced to an enormous extent. This machine facilitates to reduce one's efforts for normal daily work and helps to make it easier than ever. As our system is totally machine-driven therefore it permits the user to investigate the air quality at each moment and begin the air cleaner if the air quality isn't appropriate for breathing. If the room air quality gets higher our system can stop and help reduce electricity consumption. The foremost necessary feature of this method is that it will open the air filter and system and shut itself, this will help the air filter (HEPA) more to work more. Air is a very important source of nature, and this is particularly important for us to keep it clean, our system helps us to do the same.

REFERENCES

- [1] Indoor Air Quality Monitoring using Wireless Sensor Network Sayantani Bhattacharya, Sri Devi S, Pitchiah R Centre for Development of Advanced Computing Chennai, India.
- [2] An Intelligent and Telecontrol Environmental Monitoring Equipment with Extended Interfaces Daihong Ding, Huan Chen, Liang Zhang, Huabin Chen, Hezhi Lin, Fenglian Gao* Dept. of Automation Xiamen University Xiamen, China.
- [3] A WiFi-enabled Indoor Air Quality Monitoring and Control System: the Design and Control Experiments Xiaoke Yang, Lingyu Yang, and Jing Zhang.
- [4] Intelligent Controlling of Indoor Air Quality based on Remote Monitoring Platform by Considering Building Environment ShaodanZhi Technical Center, China State Construction Engineering Corporation Limited No. 15 of Linhe Street, Shunyi District, Beijing 101300, China.
- [5] Intelligent Controlling of Indoor Air Quality based on Remote Monitoring Platform by Considering Building Environment Yongbin Wei Technical Center, China State Construction Engineering Corporation Limited No. 15 of Linhe Street, Shunyi District, Beijing 101300, China.
- [6] Intelligent Controlling of Indoor Air Quality based on Remote Monitoring Platform by Considering Building Environment Zhen Cao Technical Center, China State Construction Engineering Corporation Limited No. 15 of Linhe Street, Shunyi District, Beijing 101300, China.
- [7] Intelligent Controlling of Indoor Air Quality based on Remote Monitoring Platform by Considering Building Environment Chao Hou Technical Center, China State Construction Engineering Corporation Limited No. 15 of Linhe Street, Shunyi District, Beijing 101300, China.
- [8] Al-Haija, Q. A., Al-Qadeeb, H., & Al-Lwaimi, A., "Case study: Monitoring of air quality inking Faisal University using a microcontroller and WSN", *Procedia Computer Science*, volume 21, pp.517–521, 31 Dec 2013.